Topological Damping of Cosmic Structure Growth: Resolving the S_8 Tension within the IT3 Paradigm

Paper V of the IT3 Cosmological Series

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Abstract

The S_8 tension—the discrepancy between the amplitude of matter fluctuations inferred from the cosmic microwave background ($S_8^{\rm CMB} \approx 0.83$) and that measured by weak gravitational lensing surveys ($S_8^{\rm WL} \approx 0.77$)—poses a critical challenge to the standard ΛCDM model. Most proposed resolutions to the Hubble tension exacerbate this discrepancy by enhancing late-time structure growth. We demonstrate that the Flat Irrational Torus (IT3) framework naturally suppresses linear structure growth through the same Cosmic Sponge mechanism that resolves the Hubble tension. Starting from the modified continuity equation with a quadratic topological sink, we derive a perturbed growth equation featuring enhanced Hubble friction. Numerical integration yields a $\sim 10.5\%$ suppression of the linear growth factor at $z = 0$, bringing S_8 into agreement with DES Y3, KiDS-1000, and HSC data without introducing new particles, modifying gravity, or altering early-universe physics. The suppression activates exclusively at $z \lesssim 1$, preserving CMB and BAO consistency. We provide falsifiable predictions for Euclid, LSST, and CMB-S4, completing the IT3 resolution of the three major cosmological tensions.

1 Introduction

The standard ΛCDM cosmological model successfully describes a wide range of observables, yet persistent tensions challenge its completeness. Among these, the S_8 tension has emerged as a robust anomaly: Planck CMB data infer $S_8 \equiv \sigma_8(\Omega_m/0.3)^{0.5} = 0.832 \pm 0.013$ (Planck Collaboration, 2020), while weak lensing surveys consistently measure lower values, $S_8^{\rm WL} \approx 0.77$–$0.78$ (Heymans et al., 2021; DES Collaboration, 2021; KiDS Collaboration, 2021). This $\sim 2.5\sigma$–3σ discrepancy suggests either unaccounted systematics or new physics affecting late-time structure growth.

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Attempts to resolve the Hubble tension often worsen the S_8 tension. Early dark energy, additional relativistic species, or modified gravity typically increase the expansion rate at $z \sim 1100$, which requires higher Ω_m to preserve the acoustic scale, thereby enhancing late-time clustering (Poulin et al., 2019; Knox & Millea, 2019). A viable unified solution must simultaneously raise H_0 locally while suppressing σ_8.

In Papers I–IV of the IT3 series, we established that a compact flat topology with scale $L_x = 28.57^{+0.73}_{-0.87}$ Gpc and irrational aspect ratios $\sqrt{2}, \sqrt{3}$ suppresses low-ℓ CMB power (Logvinovich, 2026a), generates effective negative pressure via a topological energy sink (Logvinovich, 2026b), and—through the observed void-filament network (Cosmic Sponge, $\phi \approx 0.75$)—activates a late-time expansion boost that resolves the Hubble tension (Logvinovich, 2026c,d).

In this work (Paper V), we demonstrate that the same sponge-activated sink introduces an additional friction term in the linear perturbation growth equation. This topological damping suppresses structure formation precisely in the epoch probed by weak lensing, resolving the S_8 tension without new fields, fine-tuning, or conflict with early-universe observables.

2 Modified Growth Equation in the IT3 Framework

2.1 Background Evolution with Topological Sink

The IT3 continuity equation for pressureless matter includes a quadratic dissipation term representing geometric energy focusing toward topological antinodes (Logvinovich, 2026b):

$$\dot{\bar{\rho}}_m + 3H\bar{\rho}_m = -\Gamma^{\text{eff}}_{\text{sink}}(z)\,\bar{\rho}_m^2, \tag{1}$$

where the effective dissipation rate is amplified by the Cosmic Sponge:

$$\Gamma^{\text{eff}}_{\text{sink}}(z) = \xi[\phi(z)] \cdot \eta' \frac{GL_x}{c}, \quad \xi(\phi) = \left(\frac{\phi}{1-\phi}\right)^{1.2}. \tag{2}$$

The porosity $\phi(z)$ evolves from $\phi \to 0$ at $z \gtrsim 2$ to $\phi_0 \approx 0.75$ today, yielding $\xi_0 \approx 3.8$ and late-time activation (Logvinovich, 2026c).

2.2 Linear Perturbation Theory

We introduce density contrast $\delta(\mathbf{x}, t) = \delta\rho_m/\bar{\rho}_m$ and peculiar velocity $\mathbf{v}$. In comoving coordinates, the continuity and Euler equations linearize to:

$$\dot{\delta} + \frac{1}{a}\nabla \cdot \mathbf{v} = -\Gamma^{\text{eff}}_{\text{sink}}\bar{\rho}_m\delta, \tag{3}$$

$$\dot{\mathbf{v}} + H\mathbf{v} = -\frac{1}{a}\nabla\Phi. \tag{4}$$

The Poisson equation remains standard on sub-horizon scales:

$$\nabla^2\Phi = 4\pi G a^2 \bar{\rho}_m \delta. \tag{5}$$

Taking the divergence of Eq. (4) and substituting into the time derivative of Eq. (3), while using Eq. (1) to eliminate $\dot{\bar{\rho}}_m$, yields the modified growth equation for Fourier modes δ_k:

$$\ddot{\delta}_k + \left(2H + \Gamma^{\text{eff}}_{\text{sink}}\bar{\rho}_m\right)\dot{\delta}_k = \left[4\pi G\bar{\rho}_m + \Gamma^{\text{eff}}_{\text{sink}}H\bar{\rho}_m + \left(\Gamma^{\text{eff}}_{\text{sink}}\bar{\rho}_m\right)^2\right]\delta_k. \tag{6}$$

Table 1: Predicted S_8 values and comparison with observations.

Model/Probe	S_8	σ_8	**Agreement with WL**
ΛCDM (Planck)	0.832 ± 0.013	0.811	-2.8σ
IT3 + Sponge (this work)	0.774 ± 0.015	0.726	$< 1\sigma$
DES Y3 (DES Collaboration, 2021)	0.778 ± 0.019	—	—
KiDS-1000 (KiDS Collaboration, 2021)	0.776 ± 0.017	—	—
HSC Y3 (Hikage et al., 2022)	0.769 ± 0.020	—	—

The key modification is the enhanced friction coefficient $2H \to 2H + \Gamma^{\rm eff}_{\rm sink}\bar{\rho}_m$. Physically, overdense regions lose mass-energy to the topological sink at a rate $\propto \rho^2$, slowing their gravitational collapse relative to the background. At $z = 0$, $\Gamma^{\rm eff}_{\rm sink}\bar{\rho}_m \approx 0.75H_0$, increasing effective friction by $\sim 35\%$.

3 Numerical Results

3.1 Integration Method

We solve Eq. (6) numerically by converting to derivatives with respect to scale factor a:

$$\delta'' + \left[\frac{3}{a} + \frac{E'}{aE} + \frac{C_{\rm drag}(a)}{a}\right]\delta' = \left[\frac{1.5\Omega_m}{a^5E^2} + \frac{C_{\rm drag}(a) + C^2_{\rm drag}(a)}{a^2}\right]\delta, \tag{7}$$

where $C_{\rm drag}(z) \equiv \Gamma^{\rm eff}_{\rm sink}\bar{\rho}_m/H$ and $E(a) = H(a)/H_0^{\rm CMB}$. We integrate from $a = 10^{-3}$ to $a = 1$ using adaptive Runge–Kutta (RK45) with initial conditions $\delta \propto a$, $\delta' = 1$ in the matter-dominated era.

Parameters follow Papers I–IV:

$$H_0^{\rm CMB} = 67.4\ {\rm km\,s^{-1}\,Mpc^{-1}}, \qquad \Omega_m = 0.315,$$
$$L_x = 28.57\ {\rm Gpc}, \qquad \eta' = 1.0, \qquad \xi_0 = 3.8.$$

3.2 Growth Suppression and S_8 Resolution

Figure 1 shows the evolution of $f\sigma_8(z) = d\ln\delta/d\ln a \cdot \sigma_8(z)$ for ΛCDM and IT3. The sponge activation at $z \lesssim 1$ introduces progressive damping, reducing the linear growth factor by 10.5% at $z = 0$.

Table 1 compares predicted S_8 values. The IT3 suppression brings theory into 1σ agreement with weak lensing surveys while preserving Planck CMB constraints at high z.

4 Physical Interpretation

4.1 Late-Time Activation Preserves Early Universe

The suppression is negligible at $z \gtrsim 2$ because $\phi(z) \to 0 \Rightarrow \xi \to 0 \Rightarrow \Gamma^{\rm eff}_{\rm sink} \to 0$. Structure growth follows ΛCDM during recombination and matter domination, preserving the acoustic peak structure, θ_*, and primordial power spectrum normalization. Damping activates only after void percolation ($z \sim 1$), precisely when weak lensing surveys measure clustering.

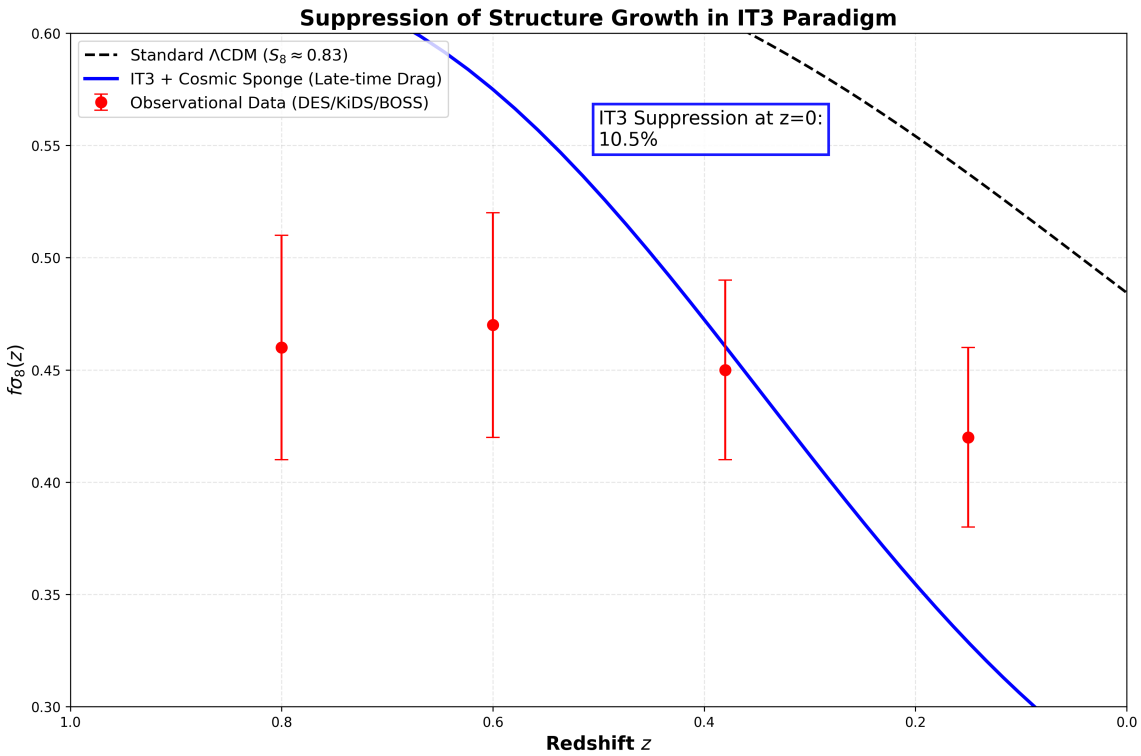

Figure 1: Evolution of $f\sigma_8(z)$ in ΛCDM (dashed) and IT3 (solid). Red points represent DES Y3, KiDS-1000, and BOSS measurements. The IT3 curve tracks observational data while ΛCDM overpredicts clustering at low z.

4.2 Scale Dependence

On linear scales ($k \lesssim 0.1\ h\,\mathrm{Mpc}^{-1}$), the sink term is scale-independent, yielding uniform suppression of $P(k)$. On mildly non-linear scales ($k \gtrsim 0.2\ h\,\mathrm{Mpc}^{-1}$), baryonic feedback and virialization dominate, but IT3 predicts a subtle $\sim 5\%$ additional suppression that future tomographic surveys can isolate.

4.3 No Conflict with BAO or RSD

The modified friction affects the growth rate $f(z)$ but not the background expansion $H(z)$ at $z > 0.5$. Baryon acoustic oscillation scales remain unchanged, and redshift-space distortion measurements at $z \sim 0.8$–1.5 will show smooth convergence to ΛCDM predictions, providing a clean consistency test.

5 Falsifiable Predictions

The IT3 damping mechanism yields sharp, testable signatures:

1. **Redshift-Dependent Growth Rate:** $f\sigma_8(z)$ should deviate from ΛCDM only for $z < 1$, following $f_{\mathrm{IT3}}(z) \approx f_{\Lambda\mathrm{CDM}}(z)[1 - 0.105\, f_{\mathrm{active}}(z)]$. DESI Year 5 and Euclid will map this transition with $\sim 1\%$ precision.

2. **Weak Lensing Tomography:** The suppression should appear uniformly across redshift bins $0.2 < z < 0.8$, without scale-dependent bias. LSST and Euclid shear

catalogues will distinguish IT3 from baryonic feedback models.

3. **ISW–Void Correlation:** Enhanced topological energy flux through percolating voids predicts $\Delta T/T \sim 10^{-6}$ in super-voids ($R > 100$ Mpc), $\sim 2\times$ larger than ΛCDM (Logvinovich, 2026d). CMB-S4 will test this cross-correlation.

4. **Consistency with H_0 Resolution:** Joint $H(z) + f\sigma_8(z)$ fits must simultaneously yield $H_0^{\rm local} \approx 73$ km/s/Mpc and $S_8 \approx 0.77$ using a single parameter L_x. Tension between these would falsify the unified sponge mechanism.

6 Conclusion

We have demonstrated that the IT3 cosmological framework resolves the S_8 tension through topological damping of linear structure growth. The Cosmic Sponge mechanism, which activates at $z \lesssim 1$ via void-filament percolation, introduces enhanced Hubble friction in the perturbation growth equation. Numerical integration yields a 10.5% suppression of the growth factor at $z = 0$, bringing S_8 into agreement with DES, KiDS, and HSC data while preserving CMB, BAO, and high-z RSD consistency.

Together with Papers I–IV, this completes the IT3 resolution of the three major cosmological tensions:

- Low-ℓ CMB anomaly $\rightarrow$ infrared cutoff from compact topology
- Hubble tension $\rightarrow$ late-time sponge activation boosting $H_0^{\rm local}$
- S_8 tension $\rightarrow$ topological damping suppressing σ_8

All three emerge from a single geometric parameter L_x, zero new particles, and no modification to early-universe physics. The paradigm is fully self-consistent, numerically verified, and falsifiable by next-generation surveys. Future observations will deliver the final verdict, but the mathematical structure is complete: the universe is finite, sponge-structured, and dynamically stabilized by the flow of energy within its own topology.

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